

torch.linalg.lu_factor_ex#

https://pytorch.org/docs/stable/generated/torch.linalg.lu_factor_ex.html

Description:

This is a version of `lu_factor()` that does not perform error checks unless `check_errors= True`. It also returns the info tensor returned by LAPACK's `getrf`. When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`. This function is “experimental” and it may change in a future PyTorch release.

Function Signature

```
torch.linalg.lu_factor_ex(A, *, pivot=True,  
check_errors=False, out=None)#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

`pivot` (bool, optional) – Whether to compute the LU decomposition with partial pivoting, or the regular LU decomposition. `pivot= False` not supported on CPU. Default: `True`. `check_errors` (bool, optional) – controls whether to check the content of infos and raise an error if it is non-zero. Default: `False`. `out` (tuple, optional) – tuple of three tensors to write the output to. Ignored if `None`. Default: `None`.

Returns:

A named tuple (LU, pivots, info).

Notes & Warnings

NOTE

Note When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`.

⚠ WARNING

Warning This function is “experimental” and it may change in a future PyTorch release.

Detailed Documentation

Documentation

This is a version of `lu_factor()` that does not perform error checks unless `check_errors=True`. It also returns the info tensor returned by LAPACK's `getrf`.

When the inputs are on a CUDA device, this function synchronizes only when `check_errors=True`.

This function is “experimental” and it may change in a future PyTorch release.

A (Tensor) – tensor of shape $(*, m, n)$ where $*$ is zero or more batch dimensions.

pivot (bool, optional) – Whether to compute the LU decomposition with partial pivoting, or the regular LU decomposition. `pivot=False` not supported on CPU. Default: True.

check_errors (bool, optional) – controls whether to check the content of infos and raise an error if it is non-zero. Default: False.

out (tuple, optional) – tuple of three tensors to write the output to. Ignored if None. Default: None.

A named tuple (LU, pivots, info).